

HDI Insight - Project Documentation

Project: A Comprehensive Measure of Well-Being (Human Development Index Prediction)

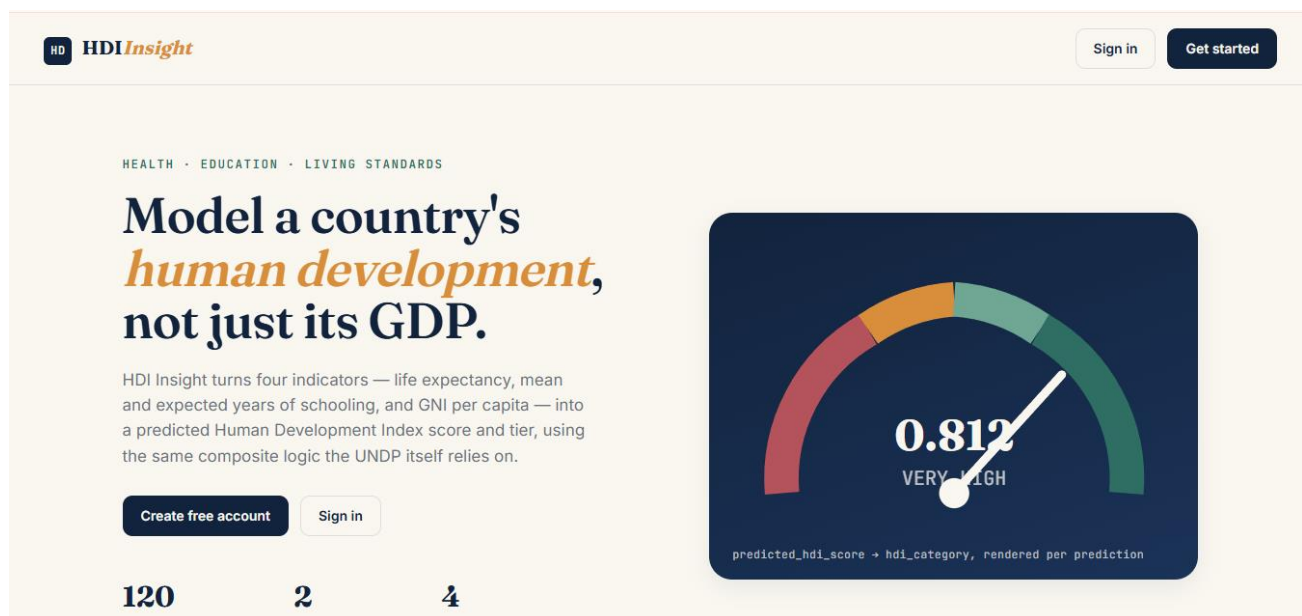
Developed using Flask, Python, Scikit-learn, SQLite, HTML, CSS, Bootstrap and Matplotlib.

1. Project Description

HDI Insight is a machine learning web application that predicts a country's Human Development Index (HDI) score and category using Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita. The application provides secure login, prediction history, country information, interactive charts, and a user-friendly dashboard.

Landing Page:

The landing page introduces HDI Insight, explains the four indicators used for prediction, and provides access to Sign In and Get Started.



Registration Page

New users create an account by entering their name, email, password, and role.

HD

HDI Insight

[Sign in](#)
[Get started](#)

Create your account

Free — used only to attribute your predictions and sessions (matches USER / SESSION in the schema).

Full name

Email

Password

Role

Researcher

Create account

[Already have an account? Sign in](#)

Prediction Page

Users select a country or enter development indicators to generate an HDI prediction.

HD

HDI Insight

[Dashboard](#)
[New Prediction](#)
[History](#)
[Countries](#)
[Model](#)

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[Sign out](#)

HOW SCORING WORKS

Your four inputs are sent to the trained RandomForest model (ML_MODEL), which returns a predicted_hdi_score. That score is bucketed into a tier using the UNDP's own thresholds:

≥ 0.800	Very High
0.700 – 0.799	High
0.550 – 0.699	Medium
< 0.550	Low

New HDI Prediction

Choose a country to auto-fill realistic starting values, then adjust as needed.

Country

India (Asia)

Life expectancy (years)

71

UNDP health dimension. Typical range 50–85.

Mean years of schooling

5.5

Average years of education completed by adults 25+.

Expected years of schooling

11.6

Years a child entering school today is expected to receive.

GNI per capita (PPP \$)

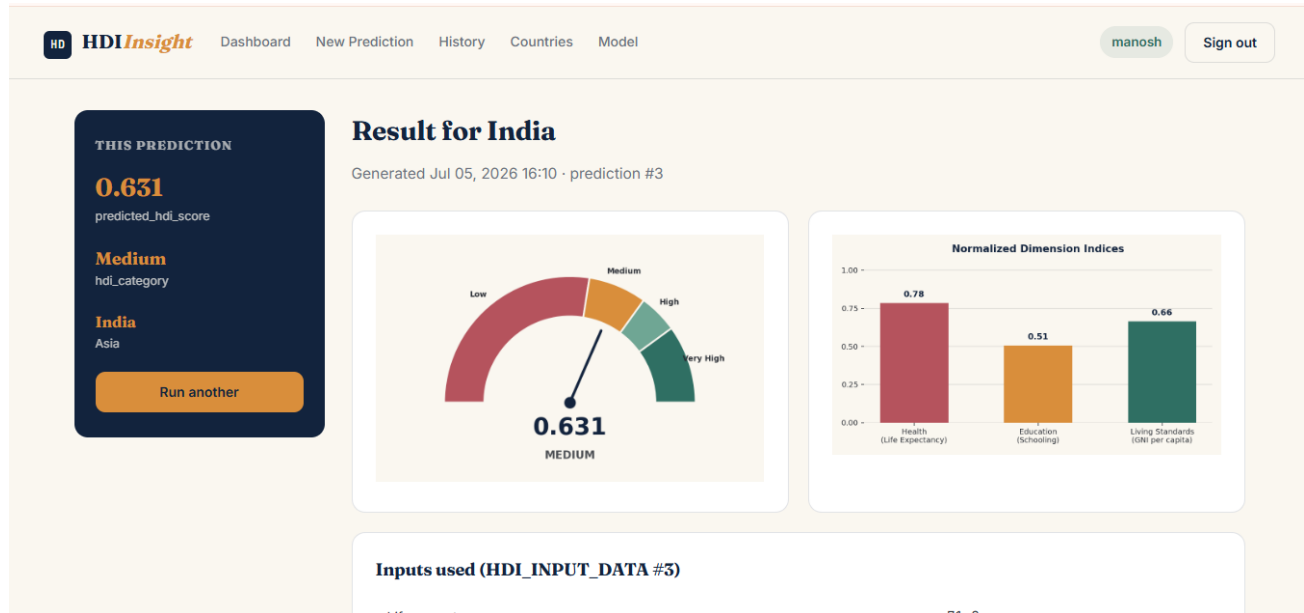
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Gross National Income per person, purchasing-power adjusted.

Run prediction

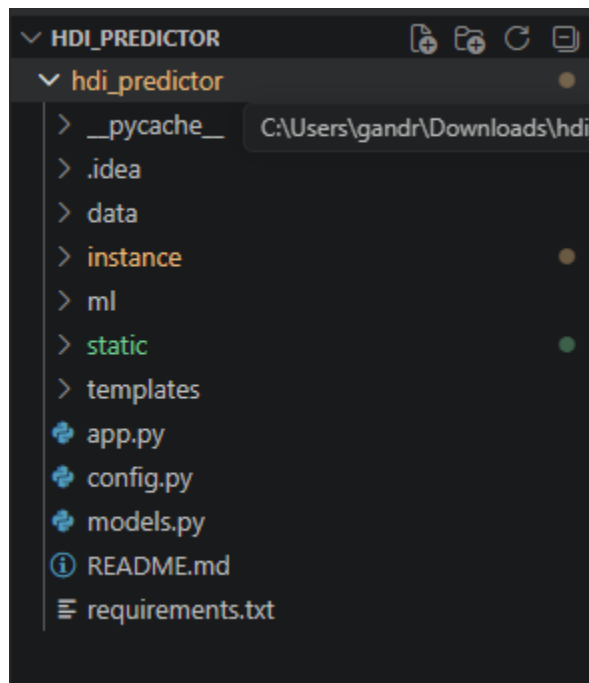
Prediction Result

Displays predicted HDI score, category, gauge visualization, normalized dimension chart, and input summary.



Project Folder Structure

Shows the Flask project organization including data, ml, templates, static, and configuration files.



Country Dataset

Country metadata including region, GNI, and HDI category used within the application.

```
COUNTRIES = [  
    ("Norway", "Europe", 5500000, "very_high"),  
    ("Switzerland", "Europe", 8800000, "very_high"),  
    ("Ireland", "Europe", 5100000, "very_high"),  
    ("Germany", "Europe", 83000000, "very_high"),  
    ("Iceland", "Europe", 380000, "very_high"),  
    ("Sweden", "Europe", 10500000, "very_high"),  
    ("Netherlands", "Europe", 17700000, "very_high"),  
    ("Denmark", "Europe", 5900000, "very_high"),  
    ("Australia", "Oceania", 26000000, "very_high"),  
    ("Finland", "Europe", 5500000, "very_high"),  
    ("Singapore", "Asia", 5900000, "very_high"),  
    ("United Kingdom", "Europe", 67500000, "very_high"),
```

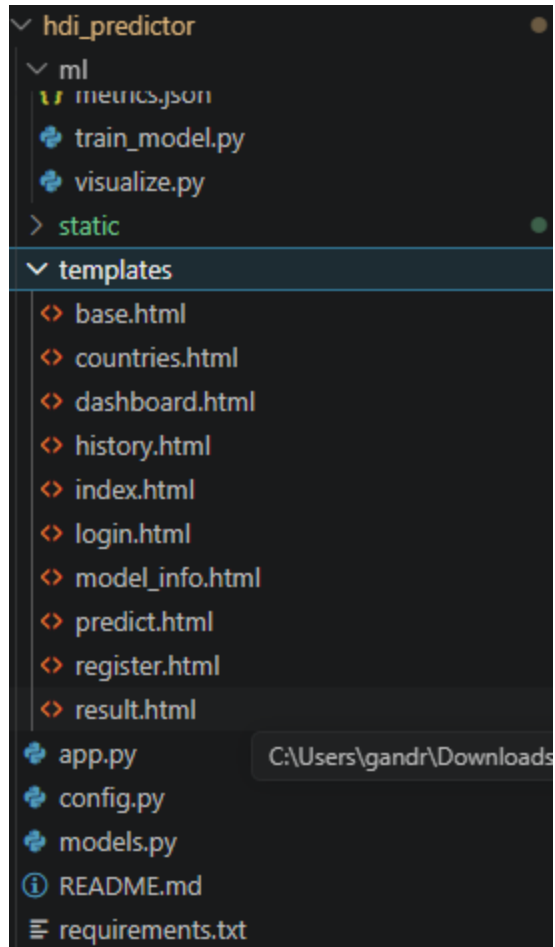
Machine Learning Training

Random Forest Regressor is trained using HDI dataset features and evaluation metrics.

```
hdi_predictor > ml > train_model.py > ...
1 import json
2 import os
3 import sys
4
5 import joblib
6 import numpy as np
7 import pandas as pd
8 from sklearn.ensemble import RandomForestRegressor
9 from sklearn.metrics import r2_score
10 from sklearn.model_selection import train_test_split
11
12 sys.path.insert(0, os.path.dirname(os.path.dirname(__file__)))
13
14 FEATURES = ["life_expectancy", "mean_years_schooling", "expected_years_s
15
16
17 def category_from_score(score):
18     if score >= 0.800:
19         return "Very High"
20     if score >= 0.700:
21         return "High"
22     if score >= 0.550:
23         return "Medium"
24     return "Low"
25
26
27 def train(dataset_path, model_out_path, metrics_out_path=None):
28     df = pd.read_csv(dataset_path)
29
30     X = df[FEATURES].values
31     y = df["hdi_score"].values
```

HTML Templates

Templates for dashboard, login, register, prediction, result, history, countries, and model pages.



Flask Backend

Main Flask application handling authentication, prediction, database interaction, and routing.

Visualization Module

Matplotlib is used to generate gauge charts and normalized dimension graphs.

```

import os
from datetime import datetime

from flask import Flask, flash, redirect, render_template, request
from flask_login import (
    LoginManager,
    current_user,
    login_required,
    login_user,
    logout_user,
)
from werkzeug.security import check_password_hash, generate_password_hash

from config import Config
from data.generate_dataset import build as build_dataset
from ml.train_model import train as train_model
from ml.visualize import generate_dimension_chart, generate_gauge_chart
from models import (
    Country,
    Dataset,
    HDIInputData,
    HDIPrediction,
    MLModel,
    Session as SessionModel,
    User,
    VisualizationReport,
    db,
)

import joblib

```

Technology Stack

Component	Technology
Frontend	HTML, CSS, Bootstrap, JavaScript
Backend	Python Flask
Machine Learning	Scikit-learn Random Forest
Database	SQLite
Visualization	Matplotlib
Version Control	Git & GitHub

Conclusion

The HDI Insight system demonstrates the application of machine learning for predicting Human Development Index values through a secure, user-friendly web interface with visualization and historical reporting.